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Rumour Clock: Visual Representation of Online Romour Spreading

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Keywords

Information Visualization; Online Rumour Spreading; RumourClock

Abstract

Information visualization has brought huge boons to multifarious areas such as sociology analysis, large hierarchies' representation, and big data demonstration. Specifically, online rumour spreading has become rampant due to the advanced development of social media. Effective governance of online rumours serves as a pivotal requirement for tycoons in social media areas like Facebook, Twitter and Instagram. However, due to the tremendous volume of everyday rumours and their dynamic essence, pure texts or simple graphs can hardly demonstrate them accurately and comprehensively. Thus, in this study, we proposed a new intuitive visualization model called RumourClock, which can handle large scale of rumours as well as compare holistically between different rumours across time.

Introduction

Information visualization (Keim, 2010) is a visual representation of information to help users understand specific knowledge or phenomena. It assists researchers because under some circumstances, visualization may reveal valuable findings behind the data. A great deal of information can be shown in a clear and organized way as well, making it easier to perceive complex issues.

Social media has promoted the dissemination of information, and becomes an ideal platform for rumour diffusing (Luo, et al., 2010). The governance of network rumours is important to social stability, economic development and national security (Zhao, et al., 2014). However, the information spreading in social media is usually large scale and social media is dynamic and complicated (Dang, et al., 2016). Related works are concentrating on analyzing who spread rumours online, why, and how. Various novel models are proposed to study the spreading mechanism of network rumours or concrete propagation relationships.

In this study, we propose RumourClock, a new intuitive visualization model, which can handle a large scale of rumours as well as compare holistically between different rumours across time.

1. Problem Statements

We have surveyed the previous studies and selected one model that is recognized as public bench marks in visualizing rumours spreading.

2.1 Reviewed Visualization Model

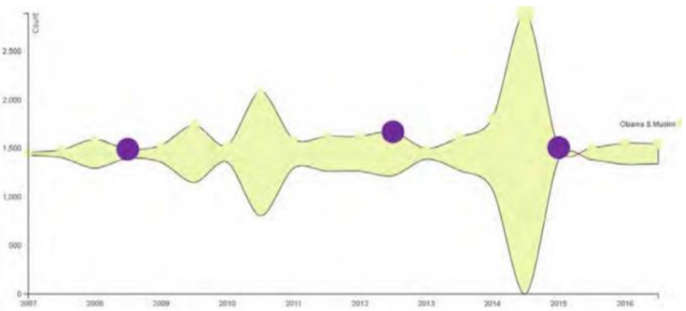

Figure 1. Visualization from " Toward Understanding How Users Respond to Rumours in Social Media"

In “Toward Understanding How Users Respond to Rumours in Social Media”, researchers used Reddit API (Weninger, Zhu & Han, 2013) and jReddit (Dehghani, Johnson, & Garten,2017) to extract 11,125 users’ comments on 195 submissions about rumour “Obama is a Muslim” (Barreto, Redlawsk, Tolbert, 2009) from Reddit since 2007 till 2015. They adopted D3 and jQUERY (Nixon, 2014) to display three types of data using multiple types of visualization methods, which were time-series graph of users’ submission, user’s interaction network graph about the selected rumour, and “who replies to whom” ground-truth user network graph. Table 7 shows the evaluation on mark and channels.

Table 1: Evaluation on the Reviewed visualization method	
Parameter	**Evaluation**
Position	Time is represented horizontally, starting from 2007 until 2015. Each node represents a submission, and the nodes are positioned according to posted time. The y-axis represents the number of comments of each submission. The purple nodes represent the submissions of one particular user’s comment.
Size	Lines indicate the general trend of the number of comments based on the rumour “Obama is a Muslim”. Normal nodes show the number of comments in each year while nodes in purple represent the submissions of one particular user.
Shape	-
Stipple/texture	-
Curvature	-
Motion	-
Title/orientation	-
Colour	Besides the basic colour of the chart, only purple was specially introduced to indicate the comments of a particular user.

Table 1. Evaluation on the Reviewed Visualization Method

This paper implements a general approach using Reddit data, and demonstrate its use by determining which users engage with a recurring rumour. It also analyzes their comments using qualitative methods. The paper seeks to reliably classify users into one of three categories: (1) “Generally support a false rumour”, (2) “Generally refute a false rumour”, or (3) “Generally joke about a false rumour”. It aims to identify and classify those rumour-spreading user categories automatically and provide a more holistic view of rumour spread in online social networks.

Table 2: Evaluation on characteristics of the Reviewed visualization method	
Characteristics	**Evaluation**
Accuracy	Due to the utilization of areas as the mark and lack of variety of visualization methods, the figure is extremely vulnerable on accuracy in demonstrating detailed and quantitative information. Readers can procure a merely abstruse understanding of changing trend of the popularity of a certain rumour. Accurate value of the number of comments in a certain year is hard to perceive.
Discriminability	The visualization in this paper seems mundane and moderate. It performs ordinarily in displaying discriminability to the users since it possesses very limited visualization elements. Only one colour is particularly exploited and all nodes are in the same size and shape.
Salience (Pop-out)	The figure is nondescript in terms of salience. It bears little clear purpose of what to present or who to present. Using purple nodes for the user “kickstand” is meaningless and elusive. Moreover, the data density is superficial, leaving the chart with huge space unexploited.
Separability	Separable representation is applied.

Grouping	Nodes in the same colour are used to display the comments of the same user. However, there may appear overlapping if more than one user comment in the same year.
Intuition	The figure is vague when it comes to intuition. The reason is it does not meet the commonsensical ways of data presentation. Users must calculate the difference between the values of the two waveforms (upper one and beneath one) to acquire the result of the number of comments in that year, which is absolutely a waste of time.
Interactivity	The interaction level is shoal. First of all, apart from the purple nodes added by the author just in order to highlight a particular user, there is no second colour in the chart except the base colour. Given that deficiency, the way users can access and interact with information are minimized. Users waste tons of time just in calculating and counting, there is no conspicuous or salient visualization to make the perception easier.
Scalability	This visualization method is poor in scalability at root. As it is mentioned above, when more users are highlighted, overlapping is ineluctable if more than one of them comment in the same year. Let alone its inborn flaw that when the user volume becomes extensive, the visualization becomes chaotic.

Table 2. Evaluation on Characteristics of the Reviewed Visualization Method

Apparently, the selected paper, namely “Toward Understanding How Users Respond to Rumours in Social Media”, has broad vacancy for us to improve its ideas and means of visualizing data, social phenomena and research findings. Hence, we further analyzed its limitations.

- Not accurate enough: Readers can procure a merely abstruse understanding of changing trend of the popularity of a certain rumour.
- Commonplace in discriminability: The chart has very limited visualization elements to present data.
- Nondescript in salience: It bears little clear purpose of what to present or who to present.
- It has potential risk in grouping: There may appear overlapping if more than one user comment in the same year.
- Intuition is diaphanous: It does not meet the commonsensical ways of data presentation. Users need to do a lot of calculation.
- Interactivity is shoal: Apart from the purple nodes added by the author just in order to highlight a particular user, there is no second colour in the chart except the base colour.
- Poor in scalability: Overlapping is a problem, while it also has its inborn flaw. When the user volume becomes extensive, the visualization becomes chaotic.

In order to improve the channel effectiveness and try to solve the current problems, we propose a new visualization method in the next chapter.

2. Research Methodology

3.1 Design Ideas

Based on the aforementioned deficiencies of Figure 1, our initial improvement was built upon the theories and concepts of radial chart, circular heatmap. The crux of our design was to improve the channel effectiveness.

First and foremost, we used 10 datasets and each contained one particular rumour with their respective number of comments. The source of the data stemmed from the same origin as our selected paper, namely Reddit. After getting the data, we pre-processed it, extracting the largest number of comments among all the 10 rumours and eliminating those rumours that had less than 100 comments. This was to exorcise anomalous data.

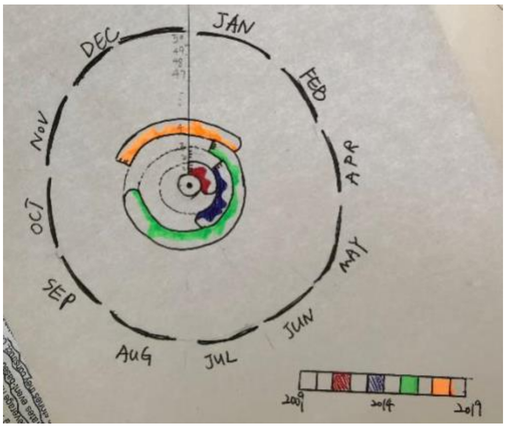

Figure 2. Design Draft of our Proposed Method RumourClock

The design was based on a circular heatmap which was partitioned into 12 sectors, representing the 12 months of a year. Each sector was segmented again into 4 small sectors, indicating the 4 weeks of a month. Afterwards, we made each sector into 10 grids in equal length. Each grid meant one rumour. The width of a single grid indicated the largest number of comments among all the 10 rumours.

Supposing the number is 3000, subsequently, we put a scale ruler and each frame counted 500. Therefore, we could easily fill up each grid which indicated each rumour. In the end, users could easily perceive a circular line on each rumour which was fluctuating according to the change of its number of comments.

What's more, since the 10 rumours might not necessarily start at the same time, we made a colour bar to as a colour collection and used different colours to represent different years from 2010 to 2018. Then we filled up each grid by referring to the colour bar and choosing proper colour. Gradient colours were used to indicate the increment of years, which was a metaphor indicating the change of years is linear.

After filling up all the grids for all 10 rumours from their respective starting point to ending point, the initial improvement design was completed. It was highly scalable which could hold more rumours just simply by adding more girds in each sector, and the unit in the scale ruler could be adjusted according to the dataset. It was also of high interactivity since there were various kinds of colours to enable users to perceive. Users were also able to zoom in to see a more detailed scale ruler and the rumour title with the number of comments at one particular time by hovering the cursor on that point. The overall picture of the design vividly demonstrated the developing trend of each rumour. Taking a closer view user could also find tons of details revealing the attributes and changes of each rumour. Hence, the new improvement was upgraded in many aspects compared to the visualization in our selected paper, including accuracy, scalability, interactivity, etc.

3.2 Development Process

Our model was based on D3 (Zhu, 2013), which is a well-known visualization tool and library. We used JavaScript and CSS files to define elements and attributes of our model while HTML files were created to visually demonstrate our model in the browser. CSV and JSON files were used to store our dataset.

We initially made a heatmap diagram (Pryke Mostaghim & Nazemi, 2007) using JavaScript, in which a new object named circosHeatmap (Zhang, Meltzer & Davis, 2013) from the class Circos was defined. In the corresponding function of circosHeatmap, 10 rumors (rumor1, rumor2 until rumor10) were created together with their respective attributes (block_id, start, end and value).

Subsequently, after finishing creating all the variables, we began to construct the layout. We drew the outermost circle representing the 12 months first. To realize the circle of months, we first needed to write code to illustrate its attributes (radius, colours, size and so on). Afterwards, we connected it to our dataset, which was a JSON file named "months". The colour of each month was derived from a JavaScript file named "colours".

After drawing the circle of months, we started to delineate the real heatmap. As it is mentioned before, in this model, we deployed 10 rumours to test our design. Therefore, we needed to portray 10 heatmap diagrams. Since we had already created 10 rumours, which meant now we can use their respective id, start time, end time and value. We made 10 heatmaps with their corresponding colours, indicating the year of each heatmap.

After all the work had been finished, we simply wrote a ".render()" to give an order to the computer and portray our tracks.

Figure 3. Visualization of the Rumours

When the heatmaps representing the years of those 10 rumours were drawn, we began to show the number of comments of each rumour on its heatmap. The way we showed each one's number of comments was by filling up the grids and in the end a circular line that was fluctuating would be created. The wavering trend of the line revealed the change of the number of comments of each rumour as time goes by.

Figure 4. Visualization of the Lines

To draw lines, we referred to our rumour dataset. We used the same source of data in our selected paper, namely Reddit. We collected and pre-processed the dataset, eliminating those anomalous data. Then we used a CSV file to store the data. Our dataset had three attributes, namely comment id, time and value. We used JavaScript to render and demonstrate the lines according to the dataset. Finally we utilized d3.queue() to load the path of our dataset.

Figure 5. Final performance of the proposed model (left) and zoom in view (right)

In the end, we rendered the heatmap diagrams and the lines together. Then we used index.html to arrange and project our final design to the browser.

3.3 Validation Test

- Accuracy. RumourClock is of high accuracy. Since we have introduced a scale ruler, and our arrangement of the heatmap is precisely according to the time duration of each rumour, one can easily read the exact number of comments of each rumour at a certain point of time. Additionally, users can see the rumour title and accurate number of comments at one particular time by hovering the cursor on that point.
- Discriminability. Firstly, the overall layout, which indicates the 12 months can be perceived. After that, number of circular heatmap diagrams representing the number of rumours can be witnessed. The colour of each heatmap showing the year of the rumour can be observed. The length of each heatmap which speaks of the duration of a particular rumour can be detected. What's more, the fluctuating line on each rumour which illustrates the change of the number of comments of a certain rumour can be noticed by the user. The scale ruler justifying the unit measurement of the number of comments can also be calculated.
- Separability. The interaction is high. We have different rumours nested inside the big circle one by one which vigorously compare their duration, changing situation of number of comments and reveal the results to the user. Multifarious colours are also deployed to vividly show different attributes to the user. Specifically, we user gradient colours to represent the years of different rumours, which in fact is a metaphor of the linear characteristic of time passing.
- Popout. In RumourClock, the peak in each rumour curve shows the abnormal data. Besides, if a user wishes to look into a particular year, the unique colour of the year is very conspicuous which renders it extremely recognizable.
- Grouping. All rumours are represented by circular heatmap diagrams, so if one wants to look for rumours, heatmap is the kind of visualization that he or she needs. All lines are used to show the number of comments of rumours and they are all arranged above the heatmaps. Therefore, it is very clear if one wants to perceive the specific value of number of comments of a rumour. This model can show the frequency of occurrence of rumours to a certain degree because users can clearly see the distribution from the visualization display.

3. Results and Discussion

Based on the evaluation test results, we proposed a new visualization design in order to improve the channel efficiency. As shown in Figure 12, there were mainly four changes, while the rest remained the same. Firstly, the original white curves were removed while each rectangular background was shaped according to the number of online rumours, which was supposed to be a better encoding with higher discriminability than the previous combination of lines and rectangles. Thirdly, when the number of online rumours was larger than two-thirds of the track height, this area was highlighted to indicate the abnormal situation. This was to achieve a higher salience, making it easier for audience to find the abnormal data. Last but not least, the colour of the scale ruler was changed to black to make it more visible.

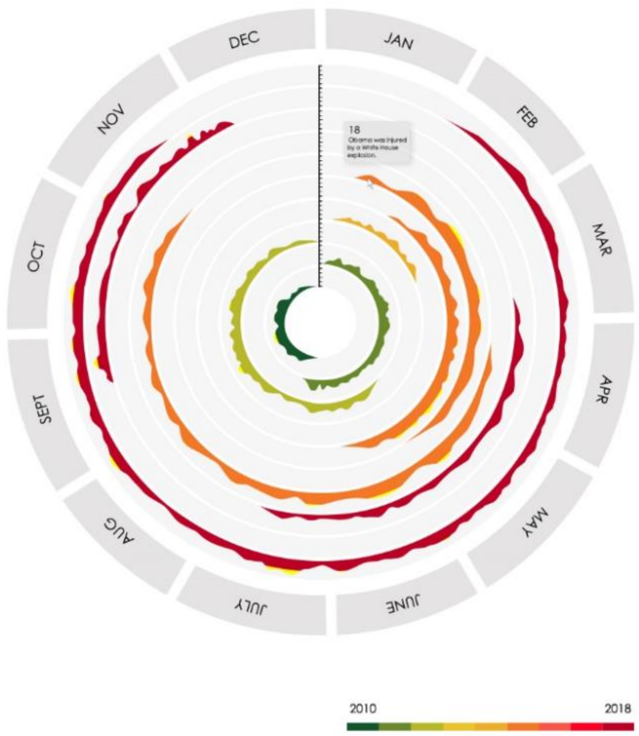

Figure 6. The newly proposed visualization method

A questionnaire survey was conducted to allow the public to evaluate our two visualization models. Participants were lecturers and undergraduate students in Xiamen University Malaysia. There were three sections in the questionnaire [8], collecting participant's basic information, feedbacks on the original model and the newly proposed model. Evaluation questions included first impression, time consumed to understand the graph, and Likert Scale questions to measure participants' opinion towards accuracy, discriminability, salience, separability, grouping, intuition, interactivity, as well as scalability.

According to the feedbacks from all the participants, although the survey was conducted separately and individually, the results were nevertheless highly consistent. Participants averagely took 1-3 minutes to understand our original model and less than 1 minute to comprehend the newly proposed model, which indicated that both models possessed high perception. The survey results also showed that when participants tested the validation of the original model for the first time, they tended to give marks between 6 and 8 (the range is 0 to 10). But when they saw our newly proposed model, the lowest mark of all the results was 8. This phenomenon showed that our first model was impressive by itself with being outperformed by our newly proposed one.

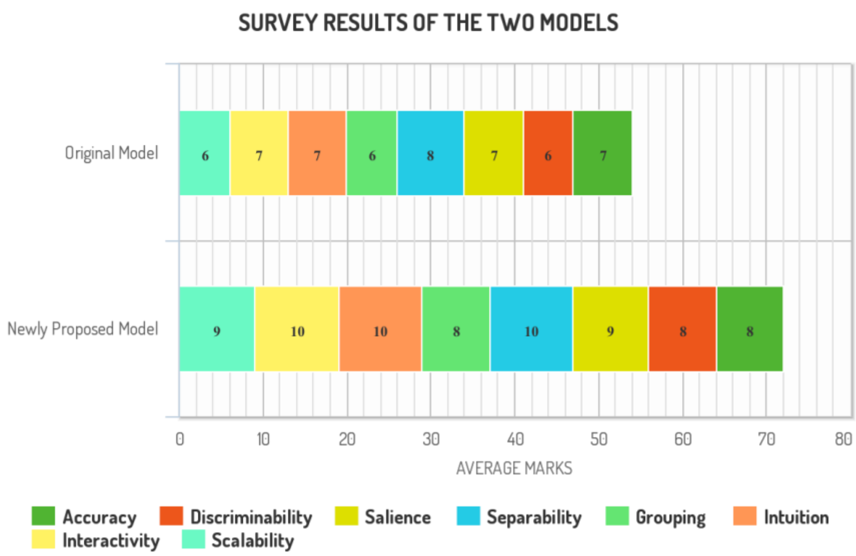

Figure 7. Survey Results of the Two Models

Therefore, considering the results of validation test and questionnaire survey, we decided that the newly proposed model should be our final design. Compared with the visualization method in our selected paper, we have done significant improvements on every dimension in terms of the validation test. For instance, the new model leveraged fluctuating background instead of the combination of lines and rectangles to represent the number of online rumours, which gave users higher discriminability.

4. Conclusion

To recapitulate, our research was to select published papers related to Information Visualization field and detect as well as ameliorate their visualization flaws. Our thesis mainly focused on online rumour spreading and we had selected one paper among three as our base model to further design our own visualization representation methods. Our final model RumourClock successfully outperformed all the baselines of the selected paper when we conducted validation test both by ourselves and by participants.

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